

INSIGHTS

PERSPECTIVES



ENVIRONMENT

Why coal ash and tailings dam disasters occur

Knowledge gaps and management shortcomings contribute to catastrophic dam failures

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On 25 January 2019, the structure damming a pond filled with iron ore mining wastes (tailings) burst at Brumadinho, Brazil (1), causing a massive mudslide that killed at least 232 people. This tailings dam failure was only the most recent in a long list of catastrophic tailings dam accidents (see the

first figure) (2, 3). Similar accidents also occur at electric power stations, where ponds are used to store coal combustion residuals such as fly and bottom ash. There are about 1000 operating ash ponds in the United States (4), and coal consumption patterns suggest that there may be more than 9000 worldwide. The catastrophic accident at the Kingston fossil power plant in Tennessee in 2008 (5) highlights the destructive potential of ash pond failures. Detailed analysis

of tailings dam and ash pond failures shows that little-understood processes such as time-delayed triggering mechanisms are more likely to manifest when best engineering practices are disregarded.

Failure of the containment structure around mine tailings and coal ash is often followed by a fast-moving mudflow, which can run downstream for several miles, with catastrophic consequences. This liquefaction of the impounded materials may sug-

Massive damage was caused by the failure of a mine tailings dam near the town of Brumadinho, Brazil. More than 232 lives were lost. Periodic failures of impounded industrial waste have catastrophic social, economic, and environmental consequences worldwide.

gest to regulators and the public that the problem lies with the impounded materials themselves. However, in the absence of internal collapse or induced shear (for example, as a result of a seismic event), liquefaction and outflow of ponded ash and tailings occur after the dam has failed. Thus, liquefaction does not cause the failure, but rather the disaster that follows.

FAILURE MECHANISMS

Forensic analyses of ash pond and tailings dam failures are hindered by the massive destruction that washes away evidence from the failed zone. Nevertheless, forensic investigations of notable dam failures have identified several mechanisms, including overtopping due to water mismanagement (Merriespruit, South Africa, 1994) (6), shear failure of foundation soils (Mount Polley, Canada, 2014) (7), and shear of compressible low-permeability tailings placed near the perimeter of the impoundment (Samarco, Brazil, 2015) (8). These postfailure investigations have often found the convergence of more than one weakness in the design, construction, and/or operation of the dam.

The Brumadinho dam was closely observed, and abundant laboratory tests, in situ tests, and monitoring data were available. Furthermore, it failed 3 years after closure, contrary to the expectation that geotechnical structures become more stable over time. Indeed, an August 2018 audit concluded that the dam was stable (9). That an apparently closely monitored and inactive dam failed catastrophically less than half a year after being audited suggests that there may be gaps in the scientific understanding of waste-storage facilities and of time-delayed triggers of failure. Filling these knowledge gaps will require changes in how coal ash and mine tailings are characterized, improved test protocols, and enhanced physics-informed data interpretation.

Tailings dams and ash ponds experience various time-dependent processes, such as gradual strains associated with pore-water pressure dissipation or creep, internal erosion and piping, and the migration of fine grains that can cause the gradual clogging of internal drainage pathways and the rise of fluid pressure within the impoundment. These processes occur throughout the entire

life of the structure, and their effects can take years or even decades to become evident. By contrast, most laboratory experiments conducted on tailings and coal ash samples last from a few hours to a couple of weeks. This discrepancy in time scales points to the need for continuous monitoring of these structures to properly assess their evolution in time.

The large mudflows that follow dam failures imply the presence of loose, water-saturated sediments that want to contract upon shear: Water cannot drain fast enough, and grains become temporarily suspended, forming a dense fluid (liquefaction). However, laboratory studies that attempt to emulate the hydraulic deposition used to deliver ash and tailings to the pond often fail to predict liquefaction under monotonic loading until impoundment depths exceed 15 to 20 m. In other words, these studies suggest that the hydraulic deposition of ash

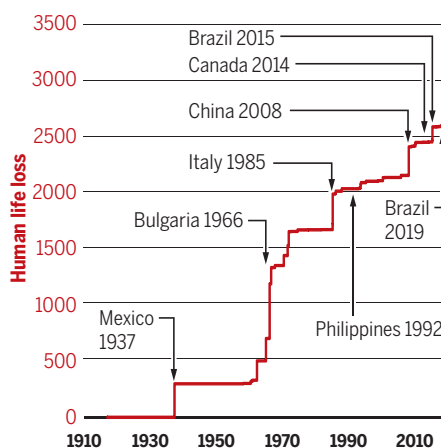
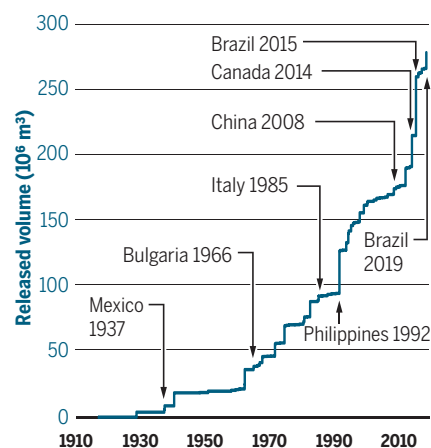
sense even minor changes in small-strain stiffness and could help to detect incipient diagenetic cementation in the field (10).

MANAGEMENT AND OPERATION

As filling of the storage pond progresses, the retaining dams for tailings or coal combustion residuals are raised by building new dykes on top of the preceding one. Each new dyke may be partially shifted toward the pond (upstream construction), aligned with the preceding dyke (centerline construction), or shifted away from the pond (downstream construction). In centerline and upstream constructions, part of the new dyke thus rests on impounded material. The upstream method (see the second figure) is the most economical, but it requires appropriate construction practices to create wide beaches made of the coarser tailings close to the dykes

Catastrophic dam failures

Over the past century, tailings dam and ash pond failures and the resulting fast-moving mudflows have led to a cumulative loss of almost 3000 lives. Data from (3).



and tailings does not necessarily produce a contractive, liquefiable sediment.

Thus, more-complex mechanisms are required to explain the time-delayed trigger and the mudflow that often follows ash ponds and tailings dam failures. These may involve coupled hydro-chemo-mechanical processes. For example, recent data show that fly ash (fine particles from burnt fuel) can experience early diagenetic cementation after placement (10). This process locks in high porosity during burial and results in a brittle material response that may favor liquefaction either by internal fabric collapse or during the shear that follows dam failure.

Cemented layers have also been identified in tailings dams (11), suggesting that mine tailings may be prone to similar effects. Shear-wave velocity measurements

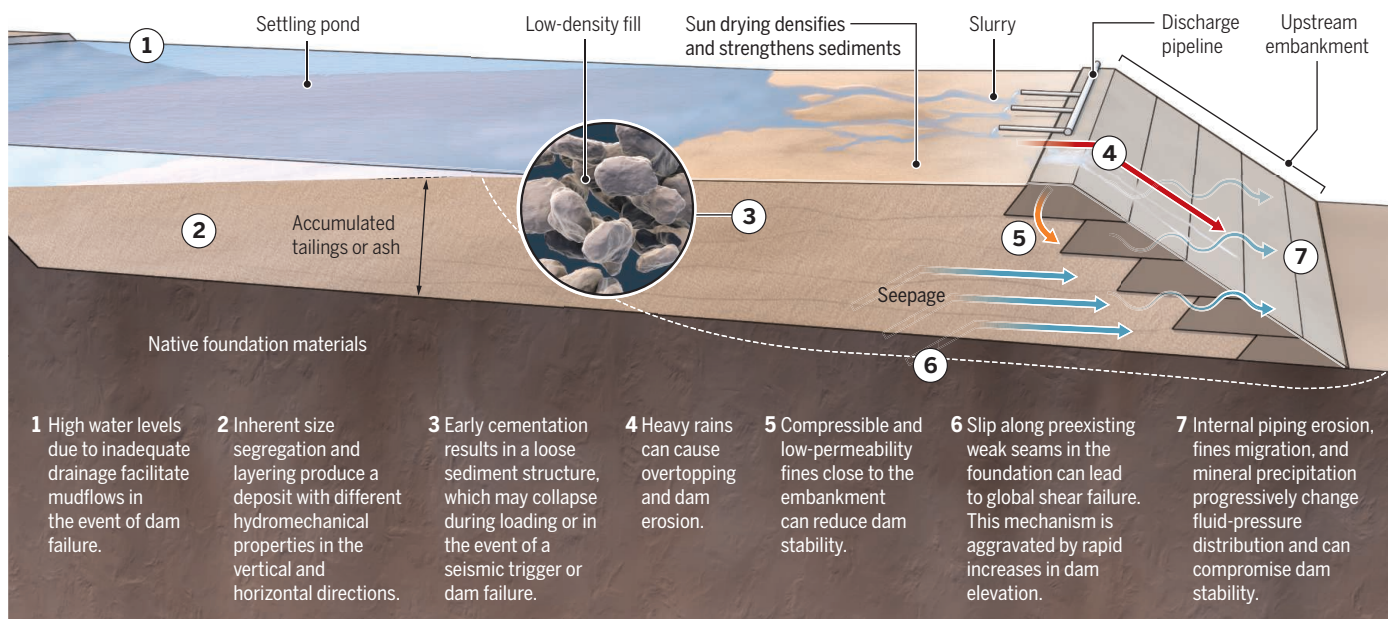
to improve drainage and stability, while the finer tailings are deposited in the pond further away from the dyke. Much of the discussion that followed the Brumadinho failure has focused on the upstream construction method used at the site (1).

Analysis of case histories points to substandard management and operational practices, often driven by the pursuit of greater financial returns, which lead mine managers to compromise on the safety of tailings dams (6). In his 1980 classical essay, the geotechnical engineer Ralph Peck noted the high probability of failure of earth dams (12). He remarked that failures were more often associated with disregard for proper engineering practices, rather than lack of knowledge. Similarly, the International Commission on Large Dams

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Mechanisms and processes of tailings dam and ash pond failures

Many different aspects of impounded ash and mine tailings and the associated dams can contribute to dam failures and the resulting fast-moving mudflows. The figure shows upstream construction as an example; most processes and mechanisms shown also apply to centerline and downstream construction methods.



investigated the failure of 221 tailings dams and also concluded that most were avoidable (13).

Indeed, postfailure investigations often highlight departures from regulation and good practice. The internal review conducted by the Inspector General of the Tennessee Valley Authority after the accident at the Kingston fossil power plant noted that the spill could be attributed to “failure of the company management to respond to a problem that was identified several years earlier” (5). The postfailure investigation of the Merriespruit tailings dam in South Africa revealed that the water level in the dam was higher than prescribed (6). The investigation of the Mount Polley tailings dam failure in Canada noted departures from approved particle size and required beach width (7). At Brumadinho, the filling of the tailings dam appears to have proceeded without clear tailings disposal guidelines between 1976 and 2005, a departure from sound engineering practice (14).

TOWARD PREVENTING DAM FAILURES

The key to understanding the cause of ash impoundment and tailings dam failures lies in identifying the triggering mechanisms that lead to the failure of the containment system. On the other hand, the key to anticipating the potential consequences of the failure lies in determining whether the impounded waste is prone to liquefaction and flow should the dam fail.

Little-understood processes such as time-delayed mechanisms are more likely

to manifest and define the response of tailings and ash impoundments when best engineering practices are disregarded. Thus, preventing failures and subsequent destructive mudflows requires enhanced physical understanding, effective engineering and management, and enforcement of regulations. Furthermore, these failures underscore the need for new performance-monitoring instrumentation, better technologies for characterizing existing impoundments, and appropriate retrofitting strategies.

Following the Brumadinho dam failure, the Brazilian authorities have banned the upstream construction method. However, most upstream dams have performed well, and storage dams built using the downstream and centerline methods have also failed. Clearly, no construction method is immune to mismanagement and poor engineering practices. In the wake of failures, special efforts should seek to identify the true triggering mechanisms and the underlying causal factors that are critical for the prevention of future accidents.

Given the number of tailings and ash impoundments around the world and their historical failure rate, more failures can be anticipated. The situation is aggravated by the tragic consequences often faced by the populations living downstream from the impoundment. Each incident prompts us to gain better physical insight, improve engineering practices, and implement regulations to minimize the potential for future catastrophes. ■

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